

PlantbreedAI Agent

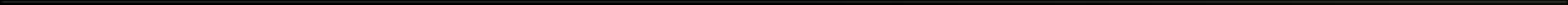
File-Level Architecture

A strategic blueprint of the modular components powering the biometrics intelligence engine.



The System Ecosystem

Mapping the end-to-end integration of frontend interface, API bridge, and scientific core logic.



Modular Entry Point



`__init__.py`

The package entry point that hoists critical classes like `PlantbreedAIAgent` up to the module level for clean, professional imports.



Public API Guard

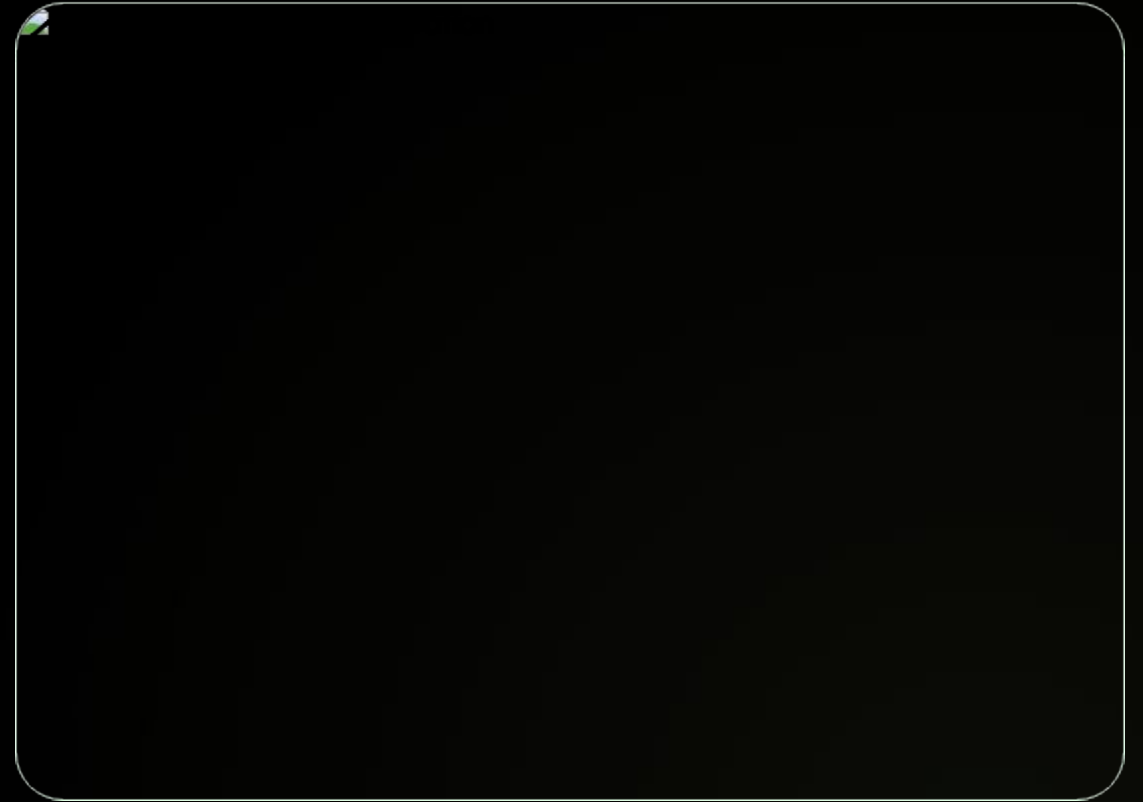
Defines the strict `__all__` list, controlling which components are visible to the rest of the application and preventing internal leaks.

Behavioral Guardrails

prompts.py

This file houses the "System Prompt"—the permanent psychological framework that defines the AI's persona, scientific precision, and strict operational rules.

Key Function: It separates the AI's "personality" from the software logic, allowing for rapid iteration of agent behavior without touching backend code.



Conversational Context



Short-Term Memory

Built on Pydantic's BaseModel to track session IDs, chat history, and active datasets with 100% data integrity.



Active Context

Remembers the "currently open" CSV and the last plot generated, enabling seamless follow-up questions without repeating details.

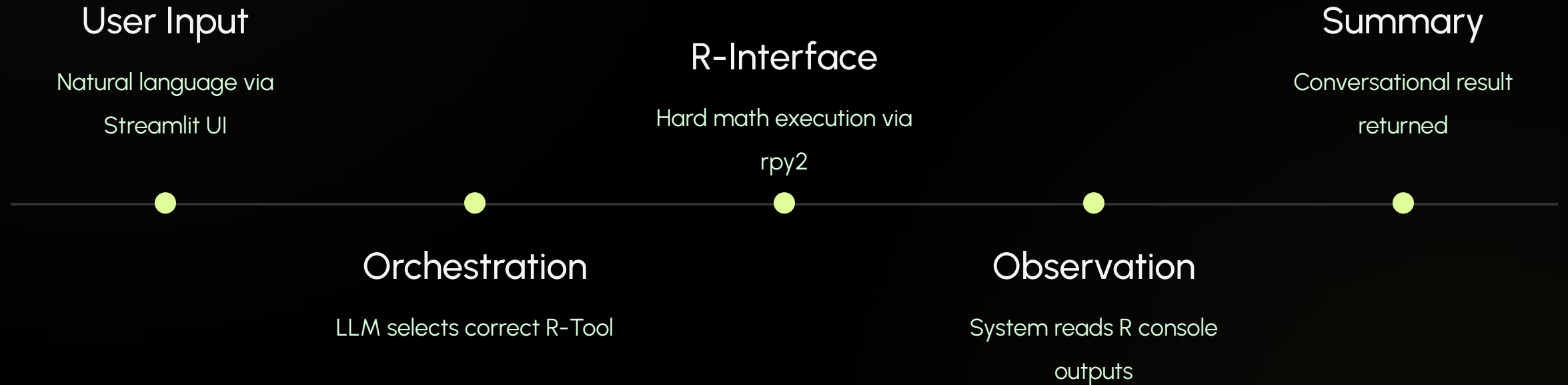
The Central Brain

`orchestrator.py`

The system's control tower. It integrates Llama 3.2 via Ollama and binds it to the R-tool suite using a "Thought-Action-Observation" loop.

It manages the deterministic execution (Temperature=0) and uses official LangChain parsers to prevent JSON hallucinations.

Logic Execution Cycle



The Physical Interface

`r_wrappers.py`

The bilingual translator bridge. It manages the absolute pathing to the scientific core logic and handles the Python-to-R data conversion.

Safety First: Implements "Safe Dev-Off" protocols to ensure the R graphics engine never crashes the backend during plot generation.



The Master Catalog



Skill Inventory

A centralized registry of all 44+ biometric skills the agent possesses, from GGE Biplots to AMMI.



Tool Discovery

Converts Python docstrings into JSON schemas that the LLM uses to "understand" which tool fits each unique request.



Extensibility

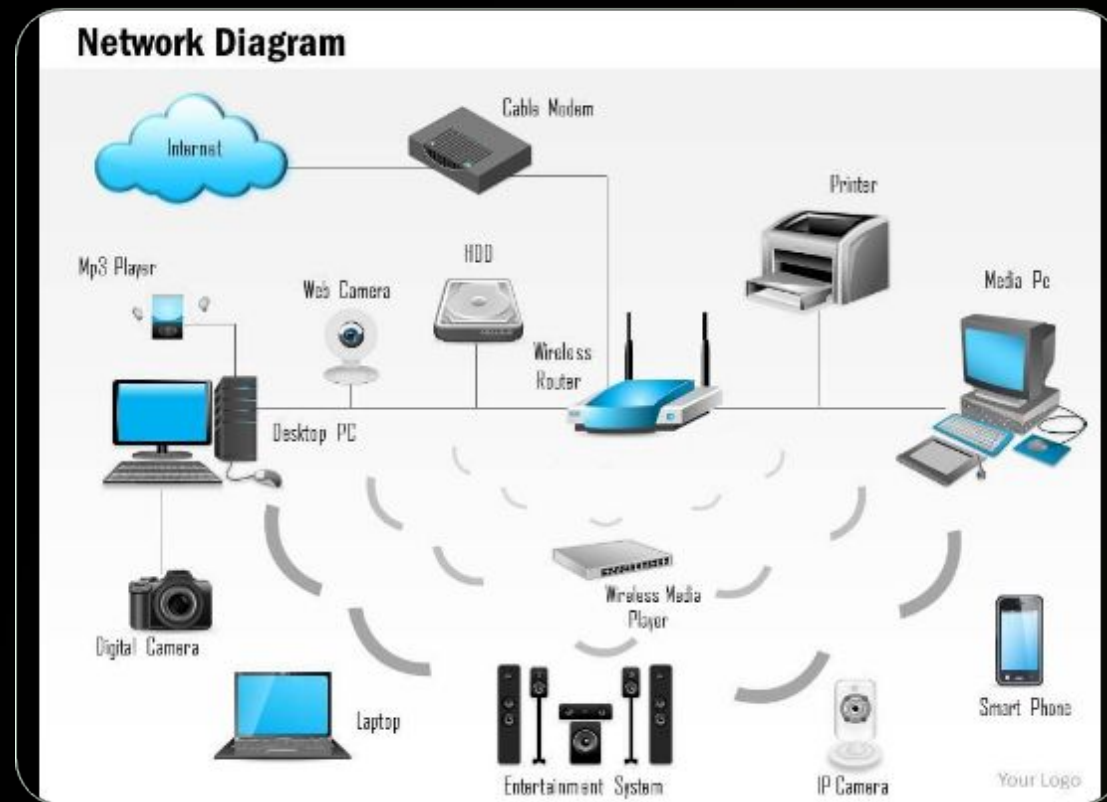
Enables adding new R-scripts to the agent's brain in seconds by simply appending to the master list.

The API Gateway

server.py

The high-speed FastAPI bridge. It acts as the "ears" of the system, listening for network requests from the web UI and routing them to the internal AI brain.

Efficiency: Initializes the heavy LLM engine once at startup, ensuring instant responsiveness for subsequent user messages.

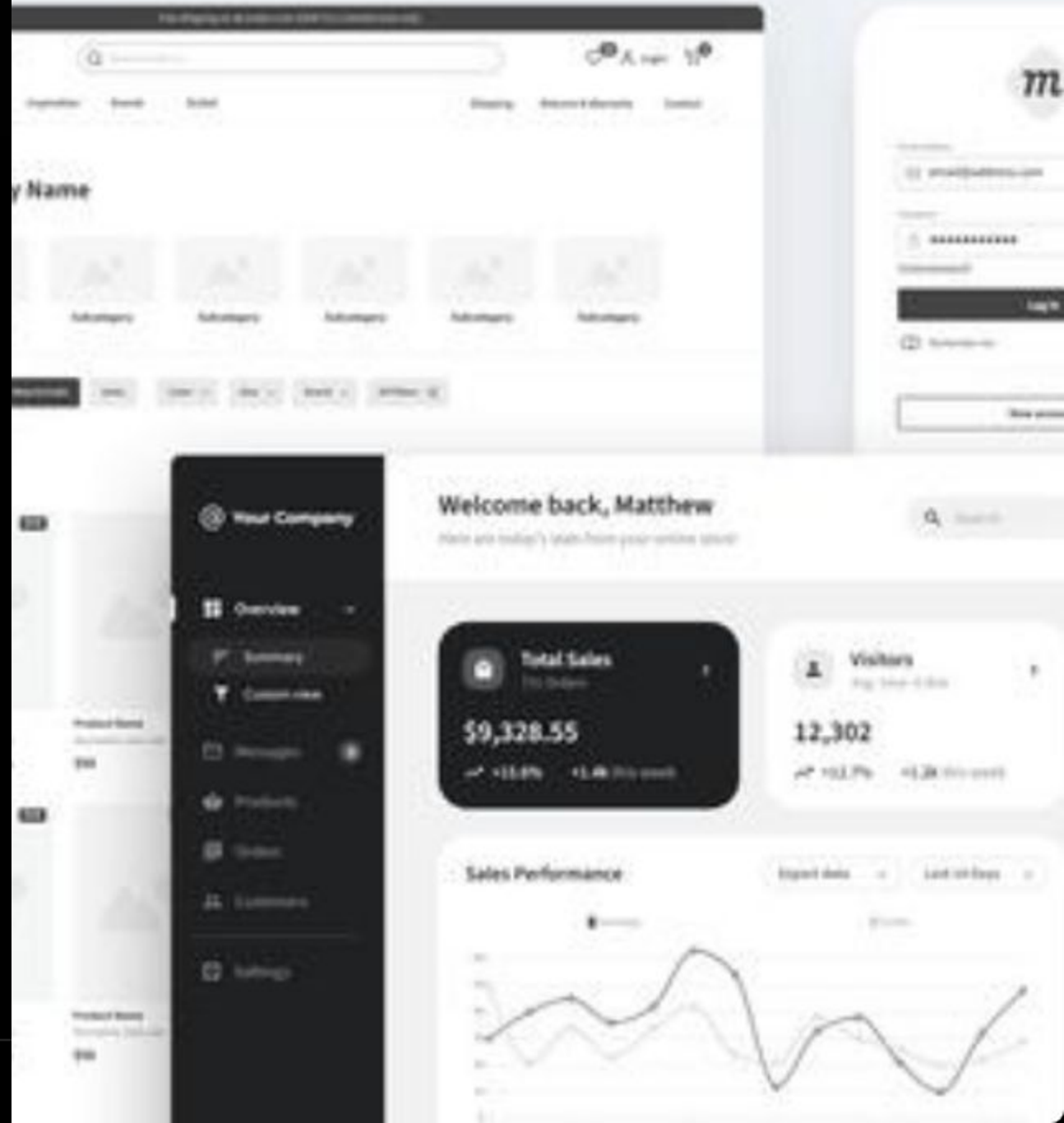


The Human Face

app.py

The interactive dashboard. Built on Streamlit to provide real-time chat bubbles, file upload capabilities, and an automated gallery of R-generated plots.

Workspace Sync: Physically writes files to the workspace inputs so the AI backend can instantly "see" uploaded data.





Operational Excellence

A unified architecture where every file serves a precise purpose in the transformation of raw field data into actionable breeding intelligence.



Image Sources



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